AMIT

AI/ML Engineer | Generative AI & RAG

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PROJECTS

Enterprise IT Support Agentic RAG Copilot | GitHub | Live Demo
Python, FastAPI, LangGraph, LangChain, ChromaDB, Groq LLM, Tavily, Docker, Render

- Built a full-stack agentic RAG IT-support copilot using LangGraph that retrieves private knowledge-base evidence, grades its sufficiency with an LLM, and falls back to optional Tavily web search with query rewrite/retry, returning source-cited answers or an explicit insufficient-evidence response.
- Engineered the retrieval and ingestion layer — persistent ChromaDB vector store with deterministic local hash embeddings (no external embedding API), LangChain loaders and recursive chunking for PDF, DOCX, TXT, and Markdown files, and an admin-key-protected upload endpoint.
- Delivered a FastAPI backend with chat, ingestion, and health endpoints, SQLite query audit logging, and a browser chat UI showing citations and workflow trace; containerized with a multi-stage, non-root Docker image and deployed on Render.

MedInsight AI — Retrieval-Augmented Generation (RAG) Medical Assistant | GitHub | Live Demo
Python, LangChain, FAISS, ONNX Runtime, Groq LLM, Streamlit

- Designed and built a Retrieval-Augmented Generation (RAG) medical assistant that ingests unstructured PDF, XML, and CSV medical records and returns context-grounded, LLM-generated answers with cited source snippets, solving the problem of scattered, hard-to-search medical documentation.
- Engineered the full retrieval pipeline — ONNX-based text embeddings (all-MiniLM-L6-v2), FAISS vector indexing, and similarity-based relevance scoring — combined with prompt engineering to feed a Groq-hosted LLM for accurate, low-latency responses.
- Delivered a production-style Streamlit application with streaming responses, session-based chat history, and chat export, while trimming heavy dependencies to keep the app lightweight for cloud deployment.

Vendor Invoice Intelligence System | GitHub | Live Demo
Python, SQL (SQLite), Scikit-Learn, Streamlit, Plotly

- Built a freight cost prediction model (Linear Regression) achieving an R^2 of 0.963 and RMSE of 124.66, outperforming Random Forest and Decision Tree baselines to automate logistics cost estimation from invoice data.
- Developed a Random Forest classification model with balanced class weights to flag high-risk invoices, joining vendor and purchase SQL tables and engineering 5 predictive features to improve audit efficiency.
- Designed SQL-driven data pipelines and an interactive Streamlit + Plotly dashboard for freight prediction, invoice risk scoring, and model performance comparison, enabling faster, data-backed audit decisions.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C++, JavaScript
Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Model Evaluation, Deep Learning Fundamentals, Scikit-learn
Generative AI & NLP: Retrieval-Augmented Generation (RAG), Agentic AI Workflows, Large Language Models (LLMs), LangChain, LangGraph, Groq API, Prompt Engineering, Natural Language Processing (NLP), Embeddings & Semantic Search, Vector Databases (FAISS, ChromaDB), Model Context Protocol (MCP)
Data Analysis & Visualization: Exploratory Data Analysis (EDA), Data Cleaning, Pandas, NumPy, Matplotlib, Plotly
Backend & APIs: FastAPI, REST APIs, Streamlit
Web Development: HTML, CSS, Tailwind CSS, React
Databases: PostgreSQL, MySQL, SQLite
DevOps & Developer Tools: Docker, Render, Git, GitHub, VS Code, Jupyter Notebook
Core CS Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA)

EDUCATION

Banaras Hindu University, Uttar Pradesh, India
MCA (Software Engineering) — CGPA: 7.98 — Aug 2024 – Dec 2026

University of Delhi, New Delhi, India
B.Sc. (Hons.) Computer Science — CGPA: 8.20 — Nov 2020 – Oct 2023

CERTIFICATIONS

- **LinkedIn Learning & Microsoft** — Career Essentials in Data Analysis
- **NPTEL (IIT Madras)** — Business Intelligence & Analytics (Elite, 84%)